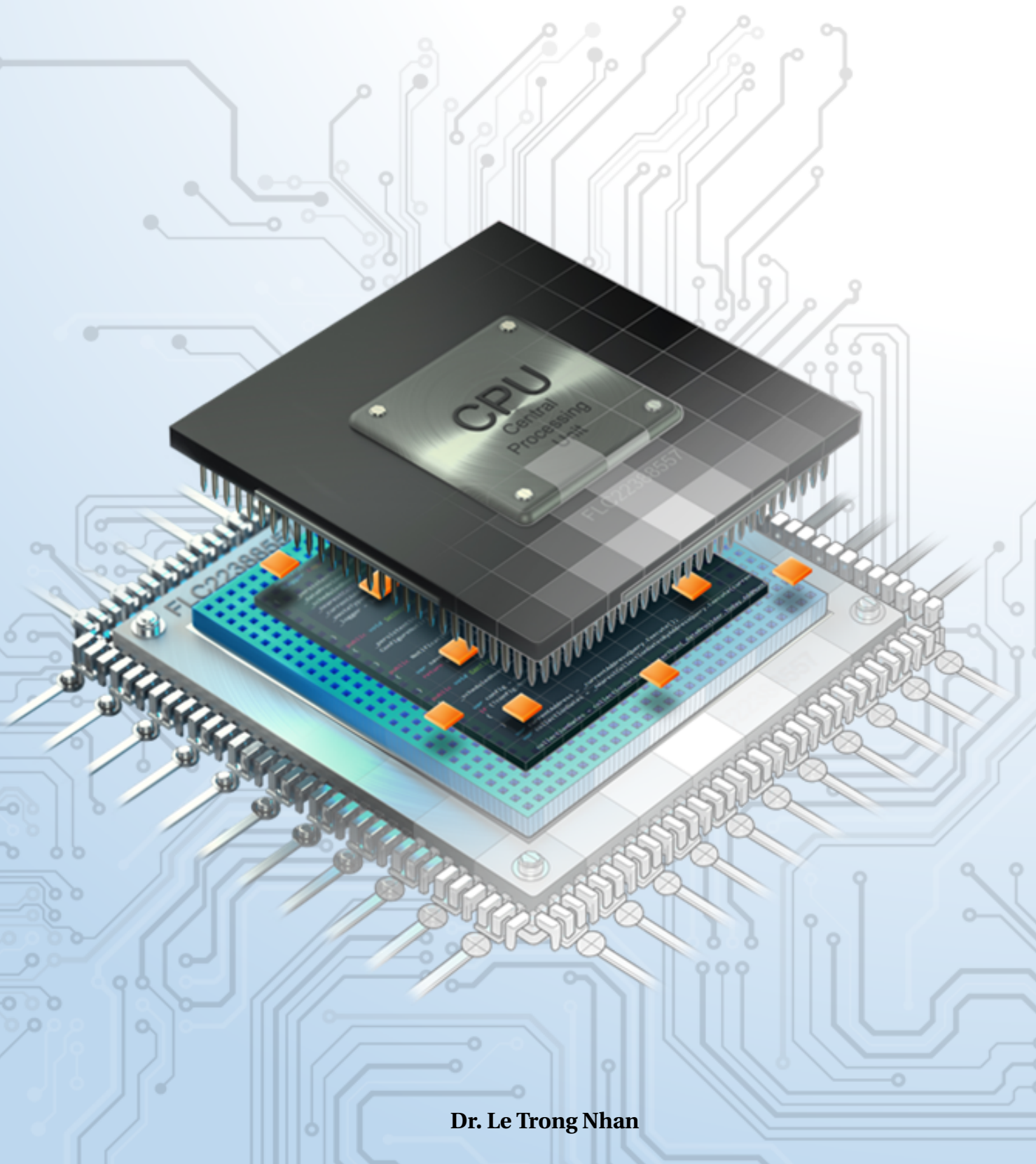




HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
COMPUTER ENGINEERING

Microcontroller



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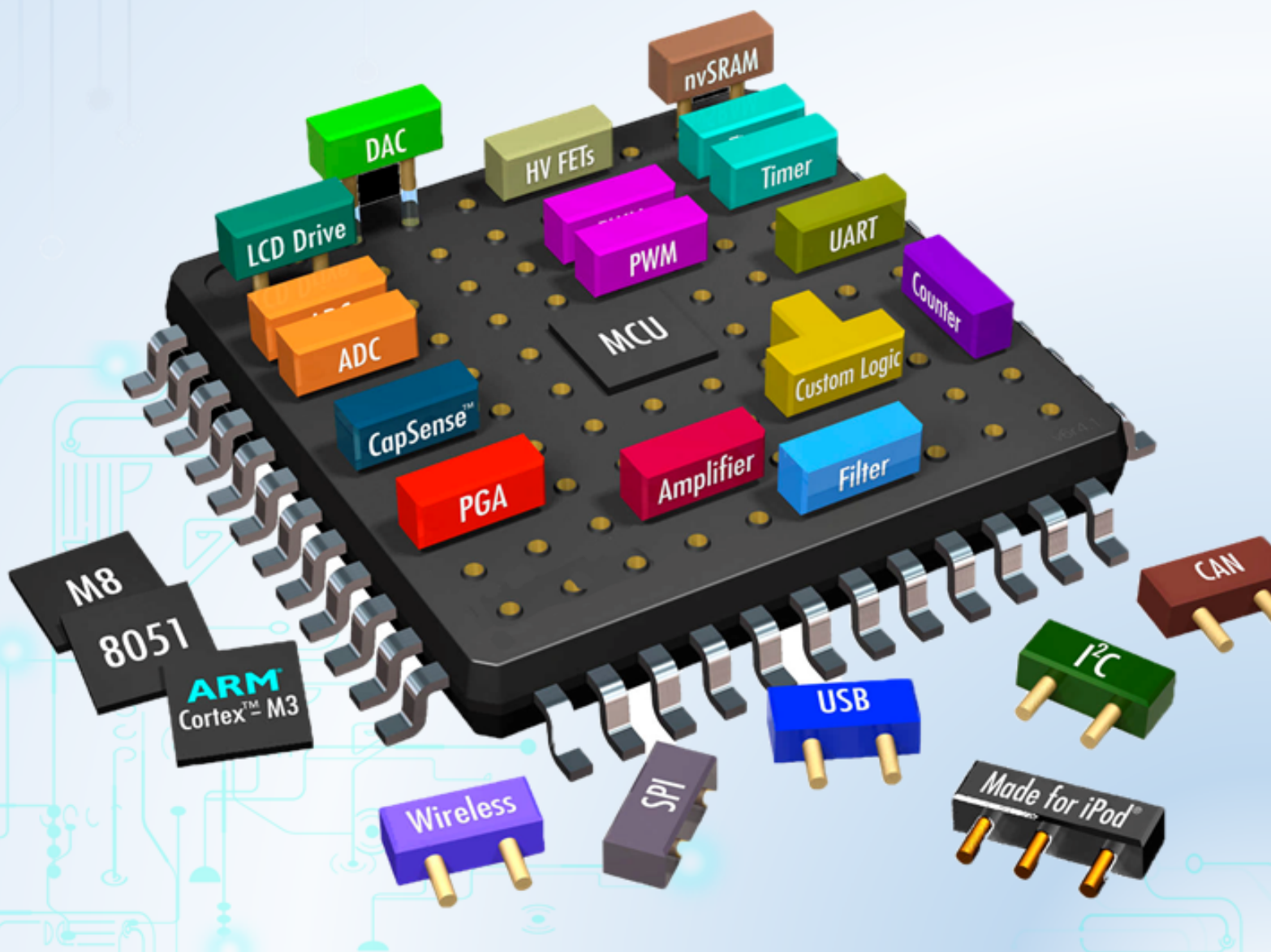
Thành phố Hồ Chí Minh, 2025

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CHƯƠNG 1

Buttons/Switches



1 Exercises and Report

1.1 Specifications

You are required to build an application of a traffic light in a cross road which includes some features as described below:

- The application has 12 LEDs including 4 red LEDs, 4 amber LEDs, 4 green LEDs.
- The application has 4 seven segment LEDs to display time with 2 for each road. The 2 seven segment LEDs will show time for each color LED corresponding to each road.
- The application has three buttons which are used
 - to select modes,
 - to modify the time for each color led on the fly, and
 - to set the chosen value.
- The application has at least 4 modes which is controlled by the first button. Mode 1 is a normal mode, while modes 2 3 4 are modification modes. You can press the first button to change the mode. Modes will change from 1 to 4 and back to 1 again.

Mode 1 - Normal mode:

- The traffic light application is running normally.

Mode 2 - Modify time duration for the red LEDs: This mode allows you to change the time duration of the red LED in the main road. The expected behaviours of this mode include:

- All single red LEDs are blinking in 2 Hz.
- Use two seven-segment LEDs to display the value.
- Use the other two seven-segment LEDs to display the mode.
- The second button is used to increase the time duration value for the red LEDs.
- The value of time duration is in a range of 1 - 99.
- The third button is used to set the value.

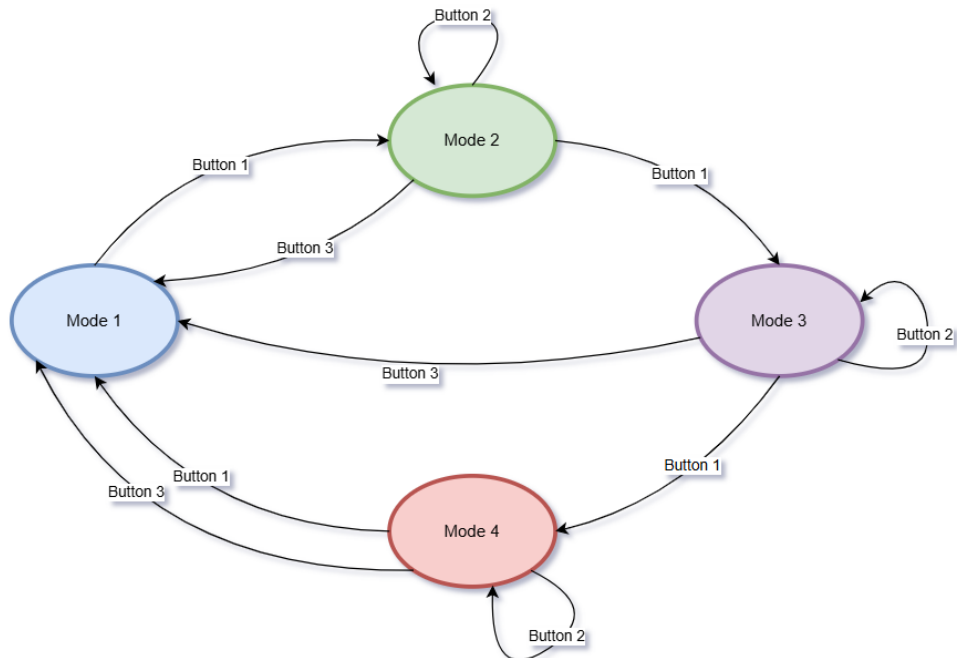
Mode 3 - Modify time duration for the amber LEDs: Similar for the red LEDs described above with the amber LEDs.

Mode 4 - Modify time duration for the green LEDs: Similar for the red LEDs described above with the green LEDs.

1.2 Exercise 1: Sketch an FSM

Your task in this exercise is to sketch an FSM that describes your idea of how to solve the problem.

Please add your report here.



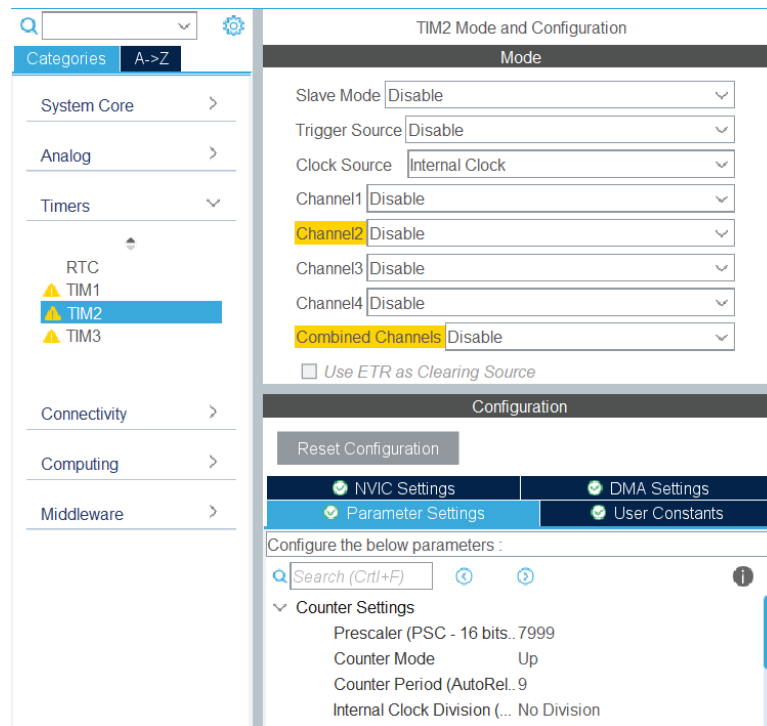
Hình 1.1: Sơ đồ máy trạng thái FSM

FSM bao gồm 4 MODE được điều khiển bởi Button 1 (nhấn Button 1 để chuyển MODE). Ở MODE 1, đèn giao thông hoạt động bình thường, với đèn đỏ sáng 5s, đèn vàng sáng 2s, đèn xanh sáng 3s. Ở MODE 2, 3, 4, người dùng có thể điều chỉnh thời gian cho từng đèn (lần lượt là đỏ, vàng, xanh) bằng Button 2 (mỗi lần nhấn tăng giá trị lên 1) và sử dụng Button 3 để lưu giá trị và quay về MODE 1.

1.3 Exercise 2: Proteus Schematic

Your task in this exercise is to draw a Proteus schematic for the problem above.

Please add your report here.



Hình 1.4: Cấu hình Timer

1.5 Exercise 4: Modify Timer Parameters

Your task in this exercise is to modify the timer settings so that when we want to change the time duration of the timer interrupt, we change it the least and it will not affect the overall system. For example, the current system we have implemented is that it can blink an LED in 2 Hz, with the timer interrupt duration is 10ms. However, when we want to change the timer interrupt duration to 1ms or 100ms, it will not affect the 2Hz blinking LED.

Please add your report here.

```

1 #ifndef INC_SOFTWARE_TIMER_H_
2 #define INC_SOFTWARE_TIMER_H_
3
4 #define MAX_TIMERS 10
5
6 void setTimer(int index, int duration);
7 int isTimerExpired(int index);
8 void timerRun();
9
10 #endif /* INC_SOFTWARE_TIMER_H_ */

```

Program 1.1: software_timer.h

```

1 #include "software_timer.h"
2
3 #define TIMER_CYCLE 10
4
5 int timer_counter[MAX_TIMERS];

```

```

6 int timer_flag[MAX_TIMERS];
7
8 void setTimer(int index, int duration) {
9     if (index >= 0 && index < MAX_TIMERS) {
10         timer_counter[index] = duration / TIMER_CYCLE;
11         timer_flag[index] = 0;
12     }
13 }
14
15 int isTimerExpired(int index) {
16     if (index >= 0 && index < MAX_TIMERS) {
17         if (timer_flag[index] == 1) {
18             timer_flag[index] = 0;
19             return 1;
20         }
21     }
22     return 0;
23 }
24
25 void timerRun() {
26     for (int i = 0; i < MAX_TIMERS; i++) {
27         if (timer_counter[i] > 0) {
28             timer_counter[i]--;
29             if (timer_counter[i] <= 0) {
30                 timer_flag[i] = 1;
31             }
32         }
33     }
34 }

```

Program 1.2: software_timer.c

```

1 #include "timer.h"
2 #include "software_timer.h"
3 #include "input_reading.h"
4 #include "led_display.h"
5 #include "main.h"
6
7 void HAL_TIM_PeriodElapsedCallback(TIM_HandleTypeDef *htim)
8 {
9     if (htim->Instance == TIM2) {
10         timerRun();
11         button_reading();
12     }
13 }

```

Program 1.3: timer.c

Giải pháp được thực hiện bằng cách xây dựng software timer. Bộ này gồm các biến `timer_counter` và `timer_flag` lưu trạng thái của từng bộ đếm. Khi hàm `setTimer` được gọi, giá trị thời gian mong muốn sẽ được quy đổi sang số chu kỳ ngắt tương

ứng bằng công thức:

$$timer_counter = \frac{duration}{TIMER_CYCLE}$$

Sau mỗi lần ngắt, hàm `timerRun` được gọi để giảm dần giá trị đếm. Khi bộ đếm về 0, cờ `timer_flag` được bật lên (bằng 1), báo hiệu thời gian đã hết.

Nhờ cơ chế này, việc thay đổi `TIMER_CYCLE` (chu kỳ ngắt phần cứng) chỉ ảnh hưởng đến tần suất gọi hàm `timerRun`, chứ không làm thay đổi logic thời gian thực của hệ thống.

1.6 Exercise 5: Adding code for button debouncing

Following the example of button reading and debouncing in the previous section, your tasks in this exercise are:

- To add new files for input reading and output display,
- To add code for button debouncing,
- To add code for increasing mode when the first button is pressed.

Please add your report here.

```
1 #include "input_reading.h"
2 #include "main.h"
3
4 #define NO_OF_BUTTONS 3
5
6 #define DURATION_FOR_AUTO_INCREASING 100
7
8 #define BUTTON_IS_PRESSED GPIO_PIN_SET
9 #define BUTTON_IS_RELEASED GPIO_PIN_RESET
10
11 static GPIO_PinState buttonBuffer[NO_OF_BUTTONS];
12
13 static GPIO_PinState debounceButtonBuffer1[NO_OF_BUTTONS];
14 static GPIO_PinState debounceButtonBuffer2[NO_OF_BUTTONS];
15
16 static uint8_t flagForButtonPress1s[NO_OF_BUTTONS];
17 static uint16_t counterForButtonPress1s[NO_OF_BUTTONS];
18
19 void button_reading(void) {
20     for (int i = 0; i < NO_OF_BUTTONS; i++) {
21         debounceButtonBuffer2[i] = debounceButtonBuffer1[i];
22     }
23
24     // Doc gia tri nut nhan
25     switch(i) {
```

```

25         case 0: // BUTTON1
26             debounceButtonBuffer1[i] = HAL_GPIO_ReadPin
(BUTTON1_GPIO_Port, BUTTON1_Pin);
27             break;
28         case 1: // BUTTON2
29             debounceButtonBuffer1[i] = HAL_GPIO_ReadPin
(BUTTON2_GPIO_Port, BUTTON2_Pin);
30             break;
31         case 2: // BUTTON3
32             debounceButtonBuffer1[i] = HAL_GPIO_ReadPin
(BUTTON3_GPIO_Port, BUTTON3_Pin);
33             break;
34         default:
35             break;
36     }
37
38     // Kiem tra tin hieu moi co bang tin hieu cu khong
39     if (debounceButtonBuffer1[i] ==
debounceButtonBuffer2[i])
40         buttonBuffer[i] = debounceButtonBuffer1[i];
41
42     if (buttonBuffer[i] == BUTTON_IS_PRESSED) {
43         // Neu mot nut duoc nhan, chung ta bat dau dem
44         if (counterForButtonPress1s[i] <
DURATION_FOR_AUTO_INCREASING) {
45             counterForButtonPress1s[i]++;
46         } else {
47             // Co se bat khi 1 giay troi qua ke tu khi
nut duoc nhsn
48             flagForButtonPress1s[i] = 1;
49         }
50     } else {
51         counterForButtonPress1s[i] = 0;
52         flagForButtonPress1s[i] = 0;
53     }
54 }
55 }
56
57 // Ham kiem tra xem nut co duoc nhan hay khong
58 unsigned char is_button_pressed(uint8_t index) {
59     if (index >= NO_OF_BUTTONS) return 0;
60     return (buttonBuffer[index] == BUTTON_IS_PRESSED);
61 }
62
63 // Ham kiem tra xem nut co duoc nhan de hay khong
64 unsigned char is_button_pressed_1s(unsigned char index) {
65     if (index >= NO_OF_BUTTONS) return 0xff;
66     return (flagForButtonPress1s[index] == 1);

```

Program 1.4: input_reading.c

```

1 // BUTTON1
2 switch(button1State) {
3     // Neu ang tha, kiem tra xem co nhan khong
4     case BUTTON_RELEASED:
5         if(is_button_pressed(0)) {
6             button1State = BUTTON_PRESSED;
7
8             // Thay doi mode
9             current_mode++;
10            if (current_mode > MODE_MODIFY_GREEN) {
11                current_mode = MODE_NORMAL;
12            }
13
14            // Kiem tra che do hien tai
15            switch (current_mode) {
16                case MODE_MODIFY_RED:
17                    temp_value = RED_DURATION;
18                    break;
19                case MODE_MODIFY_AMBER:
20                    temp_value = AMBER_DURATION;
21                    break;
22                case MODE_MODIFY_GREEN:
23                    temp_value = GREEN_DURATION;
24                    break;
25                case MODE_NORMAL:
26                    traffic_state = RED_GREEN;
27                    counter1 = RED_DURATION;
28                    counter2 = GREEN_DURATION;
29                    setTimeout(0, 1000);
30                    break;
31            }
32        }
33        break;
34
35    // Neu dang nhan, kiem tra co tha ra hay nhan giu
36    case BUTTON_PRESSED:
37        if(!is_button_pressed(0)) {
38            button1State = BUTTON_RELEASED;
39        } else {
40            if(is_button_pressed_1s(0)) {
41                button1State =
42                BUTTON_PRESSED_MORE_THAN_1_SECOND;
43            }
44            break;
45

```

```

46 // Neu dang nhan giu, kiem tra xem co tha ra khong
47 case BUTTON_PRESSED_MORE_THAN_1_SECOND:
48     if(!is_button_pressed(0)) {
49         button1State = BUTTON_RELEASED;
50     }
51     break;
52 }

```

Program 1.5: BUTTON 1 để chuyển đổi MODE

1.7 Exercise 6: Adding code for displaying modes

Your tasks in this exercise are:

- To add code for display mode on seven-segment LEDs, and
- To add code for blinking LEDs depending on the mode that is selected.

Please add your report here.

```

1 void display7SEG() {
2     for (int i = 0; i < 4; i++){
3         if (led_buffer[i] < 0 || led_buffer[i] > 9) led_buffer[
4             i] = 0;
5     }
6     HAL_GPIO_WritePin(BCD11_GPIO_Port, BCD11_Pin, (led_buffer
7         [0] & 0x01) ? GPIO_PIN_SET : GPIO_PIN_RESET); // D0
8     HAL_GPIO_WritePin(BCD12_GPIO_Port, BCD12_Pin, (led_buffer
9         [0] & 0x02) ? GPIO_PIN_SET : GPIO_PIN_RESET); // D1
10    HAL_GPIO_WritePin(BCD13_GPIO_Port, BCD13_Pin, (led_buffer
11        [0] & 0x04) ? GPIO_PIN_SET : GPIO_PIN_RESET); // D2
12    HAL_GPIO_WritePin(BCD14_GPIO_Port, BCD14_Pin, (led_buffer
13        [0] & 0x08) ? GPIO_PIN_SET : GPIO_PIN_RESET); // D3
14
15    HAL_GPIO_WritePin(BCD21_GPIO_Port, BCD21_Pin, (led_buffer
16        [1] & 0x01) ? GPIO_PIN_SET : GPIO_PIN_RESET); // D0
17    HAL_GPIO_WritePin(BCD22_GPIO_Port, BCD22_Pin, (led_buffer
18        [1] & 0x02) ? GPIO_PIN_SET : GPIO_PIN_RESET); // D1
19    HAL_GPIO_WritePin(BCD23_GPIO_Port, BCD23_Pin, (led_buffer
20        [1] & 0x04) ? GPIO_PIN_SET : GPIO_PIN_RESET); // D2
21    HAL_GPIO_WritePin(BCD24_GPIO_Port, BCD24_Pin, (led_buffer
22        [1] & 0x08) ? GPIO_PIN_SET : GPIO_PIN_RESET); // D3
23
24    HAL_GPIO_WritePin(BCD31_GPIO_Port, BCD31_Pin, (led_buffer
25        [2] & 0x01) ? GPIO_PIN_SET : GPIO_PIN_RESET); // D0
26    HAL_GPIO_WritePin(BCD32_GPIO_Port, BCD32_Pin, (led_buffer
27        [2] & 0x02) ? GPIO_PIN_SET : GPIO_PIN_RESET); // D1
28    HAL_GPIO_WritePin(BCD33_GPIO_Port, BCD33_Pin, (led_buffer
29        [2] & 0x04) ? GPIO_PIN_SET : GPIO_PIN_RESET); // D2

```



```

18 HAL_GPIO_WritePin(BCD34_GPIO_Port, BCD34_Pin, (led_buffer
    [2] & 0x08) ? GPIO_PIN_SET : GPIO_PIN_RESET); // D3
19
20 HAL_GPIO_WritePin(BCD41_GPIO_Port, BCD41_Pin, (led_buffer
    [3] & 0x01) ? GPIO_PIN_SET : GPIO_PIN_RESET); // D0
21 HAL_GPIO_WritePin(BCD42_GPIO_Port, BCD42_Pin, (led_buffer
    [3] & 0x02) ? GPIO_PIN_SET : GPIO_PIN_RESET); // D1
22 HAL_GPIO_WritePin(BCD43_GPIO_Port, BCD43_Pin, (led_buffer
    [3] & 0x04) ? GPIO_PIN_SET : GPIO_PIN_RESET); // D2
23 HAL_GPIO_WritePin(BCD44_GPIO_Port, BCD44_Pin, (led_buffer
    [3] & 0x08) ? GPIO_PIN_SET : GPIO_PIN_RESET); // D3
24
25 }
26
27 // FSM cho che do normal
28 void fsm_normal_mode(void) {
29     if (isTimerExpired(0)) {
30         setTimer(0, 1000);
31         // Cap nhat gia tri hien thi
32         led_buffer[0] = counter1 / 10;
33         led_buffer[1] = counter1 % 10;
34         led_buffer[2] = counter2 / 10;
35         led_buffer[3] = counter2 % 10;
36
37         switch (traffic_state) {
38
39             // Road 1 Do - Road 2 Xanh
40             case RED_GREEN:
41                 // Bat cac LED don
42                 setTrafficLight(0, 1, 0, 0);
43                 setTrafficLight(1, 0, 0, 1);
44
45                 counter1--;
46                 counter2--;
47                 // Het thoi gian chuyen qua trsng thai tiep theo
48                 if (counter2 <= 0) {
49                     traffic_state = RED_AMBER;
50                     counter2 = AMBER_DURATION;
51                 }
52                 break;
53
54             // Road 1 Do - Road 2 Vang
55             case RED_AMBER:
56                 setTrafficLight(0, 1, 0, 0);
57                 setTrafficLight(1, 0, 1, 0);
58
59                 counter1--;
60                 counter2--;
61                 if (counter2 <= 0) {

```

```

62     traffic_state = GREEN_RED;
63     counter1 = GREEN_DURATION;
64     counter2 = RED_DURATION;
65 }
66 break;
67
68 // Road 1 Xanh - Road 2 Do
69 case GREEN_RED:
70     setTrafficLight(0, 0, 0, 1);
71     setTrafficLight(1, 1, 0, 0);
72
73     counter1--;
74     counter2--;
75     if (counter1 <= 0) {
76         traffic_state = AMBER_RED;
77         counter1 = AMBER_DURATION;
78     }
79     break;
80
81 // Road 1 Vang - Road 2 Do
82 case AMBER_RED:
83     setTrafficLight(0, 0, 1, 0);
84     setTrafficLight(1, 1, 0, 0);
85
86     counter1--;
87     counter2--;
88     if (counter1 <= 0) {
89         traffic_state = RED_GREEN;
90         counter1 = RED_DURATION;
91         counter2 = GREEN_DURATION;
92     }
93     break;
94 }
95 }
96 }

```

Program 1.6: Hiển thị chế độ trên LED 7 đoạn

```

1 // FSM cho cac che do sua doi
2 void fsm_modify_mode(void) {
3     // Nhay LED moi 2Hz
4     if (isTimerExpired(1)) {
5         setTimer(1, 250);
6         blink_state = !blink_state;
7
8         if (blink_state) {
9             switch (current_mode) {
10                 case MODE_MODIFY_RED:
11                     // Nhay tat ca den do
12                     HAL_GPIO_WritePin(RED1_GPIO_Port,

```

```

13     RED1_Pin, GPIO_PIN_SET);
        HAL_GPIO_WritePin(RED2_GPIO_Port,
RED2_Pin, GPIO_PIN_SET);
14         break;
15     case MODE_MODIFY_AMBER:
16         // Nhảy tắt cả đèn vàng
17         HAL_GPIO_WritePin(AMBER1_GPIO_Port,
AMBER1_Pin, GPIO_PIN_SET);
18         HAL_GPIO_WritePin(AMBER2_GPIO_Port,
AMBER2_Pin, GPIO_PIN_SET);
19         break;
20     case MODE_MODIFY_GREEN:
21         // Nhảy tắt cả đèn xanh
22         HAL_GPIO_WritePin(GREEN1_GPIO_Port,
GREEN1_Pin, GPIO_PIN_SET);
23         HAL_GPIO_WritePin(GREEN2_GPIO_Port,
GREEN2_Pin, GPIO_PIN_SET);
24         break;
25     }
26     } else {
27         clearAllTrafficLights();
28     }
29 }
30
31 // Cập nhật bộ đếm hiển thị
32 led_buffer[0] = temp_value / 10;
33 led_buffer[1] = temp_value % 10;
34 // Hiển thị số thứ tự chế độ
35 led_buffer[2] = 0;
36 led_buffer[3] = current_mode;
37 }

```

Program 1.7: Hiển thị đèn LED nhấp nháy tại chế độ sửa đổi

1.8 Exercise 7: Adding code for increasing time duration value for the red LEDs

Your tasks in this exercise are:

- to use the second button to increase the time duration value of the red LEDs
- to use the third button to set the value for the red LEDs.

Please add your report here.

```

1 // BUTTON2
2 if (current_mode != MODE_NORMAL) {
3     switch(button2State) {
4         case BUTTON_RELEASED:

```

```

5         if(is_button_pressed(1)) {
6             button2State = BUTTON_PRESSED;
7             // Tang gia tri thoi gian hien thi
8             temp_value++;
9             if (temp_value > 99) {
10                 temp_value = 1;
11             }
12         }
13         break;
14
15     case BUTTON_PRESSED:
16         if(!is_button_pressed(1)) {
17             button2State = BUTTON_RELEASED;
18         } else {
19             if(is_button_pressed_1s(1)) {
20                 button2State =
21                 BUTTON_PRESSED_MORE_THAN_1_SECOND;
22             }
23             break;
24
25     case BUTTON_PRESSED_MORE_THAN_1_SECOND:
26         if(!is_button_pressed(1)) {
27             button2State = BUTTON_RELEASED;
28         } else {
29             // Neu nhan giu tang gia tri moi 500ms
30             if (isTimerExpired(2)) {
31                 temp_value++;
32                 if (temp_value > 99) temp_value = 1;
33                 setTimer(2, 500);
34             }
35         }
36         break;
37     }
38 }

```

Program 1.8: BUTTON 2 để tăng giá trị thời gian

```

1 // BUTTON3
2 if (current_mode != MODE_NORMAL) {
3     switch(button3State) {
4         case BUTTON_RELEASED:
5             if(is_button_pressed(2)) {
6                 button3State = BUTTON_PRESSED;
7
8                 // Set gia tri
9                 switch (current_mode) {
10                     case MODE_MODIFY_RED:
11                         RED_DURATION = temp_value;
12                     break;

```

```

13         case MODE_MODIFY_AMBER:
14             AMBER_DURATION = temp_value;
15             break;
16         case MODE_MODIFY_GREEN:
17             GREEN_DURATION = temp_value;
18             break;
19     }
20
21     // Kiem tra logic
22     adjustTrafficTiming();
23
24     // Tro ve che do binh thuong sau khi thiet
lap
25     current_mode = MODE_NORMAL;
26     traffic_state = RED_GREEN;
27     counter1 = RED_DURATION;
28     counter2 = GREEN_DURATION;
29     setTimer(0, 1000);
30 }
31 break;
32
33 case BUTTON_PRESSED:
34     if(!is_button_pressed(2)) {
35         button3State = BUTTON_RELEASED;
36     } else {
37         if(is_button_pressed_1s(2)) {
38             button3State =
BUTTON_PRESSED_MORE_THAN_1_SECOND;
39         }
40     }
41     break;
42
43 case BUTTON_PRESSED_MORE_THAN_1_SECOND:
44     if(!is_button_pressed(2)) {
45         button3State = BUTTON_RELEASED;
46     }
47     break;
48 }
49 }

```

Program 1.9: BUTTON 3 để lưu giá trị thời gian

```

1 // Ham kiem tra v dieu chinh logic thoi gian
2 void adjustTrafficTiming(void) {
3
4     // Quy luat logic: RED = GREEN + AMBER
5     int expected_red = GREEN_DURATION + AMBER_DURATION;
6
7     if (RED_DURATION != expected_red) {
8         if (current_mode == MODE_MODIFY_RED) {

```

```

9      // Chinh den DO      tu can doi den XANH
10     GREEN_DURATION = RED_DURATION - AMBER_DURATION;
11     if (GREEN_DURATION < 2) GREEN_DURATION = 2;
12 }
13 else if (current_mode == MODE_MODIFY_GREEN) {
14     // Chinh den XANH      tu can doi den DO
15     RED_DURATION = GREEN_DURATION + AMBER_DURATION;
16 }
17 else if (current_mode == MODE_MODIFY_AMBER) {
18     // Chinh den VANG      tu can doi den FO
19     RED_DURATION = GREEN_DURATION + AMBER_DURATION;
20 }
21 }
22
23 // Dong bo lai bien hien thi
24 counter1 = RED_DURATION;
25 counter2 = GREEN_DURATION;
26 }

```

Program 1.10: Hàm adjustTrafficTiming để kiểm tra và điều chỉnh logic thời gian

Link mô phỏng: https://drive.google.com/file/d/1D5znzrT7gupPErbYENbJGyMni5G7ipz0/view?usp=drive_link

1.9 Exercise 8: Adding code for increasing time duration value for the amber LEDs

Your tasks in this exercise are:

- to use the second button to increase the time duration value of the amber LEDs
- to use the third button to set the value for the amber LEDs.

Please add your report here.]

Đã thực hiện code ở Exercise 7.

Link mô phỏng: https://drive.google.com/file/d/1do3SbJuryPGH8LIH_F-drbs0H3m0CIQ8/view?usp=sharing

1.10 Exercise 9: Adding code for increasing time duration value for the green LEDs

Your tasks in this exercise are:

- to use the second button to increase the time duration value of the green LEDs

- to use the third button to set the value for the green LEDs.

Please add your report here.

Đã thực hiện code ở Exercise 7.

Link mô phỏng: <https://drive.google.com/file/d/1K8EHw4Rkl1NwnSM3JzxJUTiRinqo6DDo/view?usp=sharing>

1.11 Exercise 10: To finish the project

Your tasks in this exercise are:

- To integrate all the previous tasks to one final project
- To create a video to show all features in the specification
- To add a report to describe your solution for each exercise.
- To submit your report and code on the BKeL

Link mô phỏng: <https://drive.google.com/file/d/1Iix2whbuMed6j3miu3Ah4dJGnDOB9TL5/view?usp=sharing>

GitHub link for the project: <https://github.com/anhthu057/VXL-LAB3>